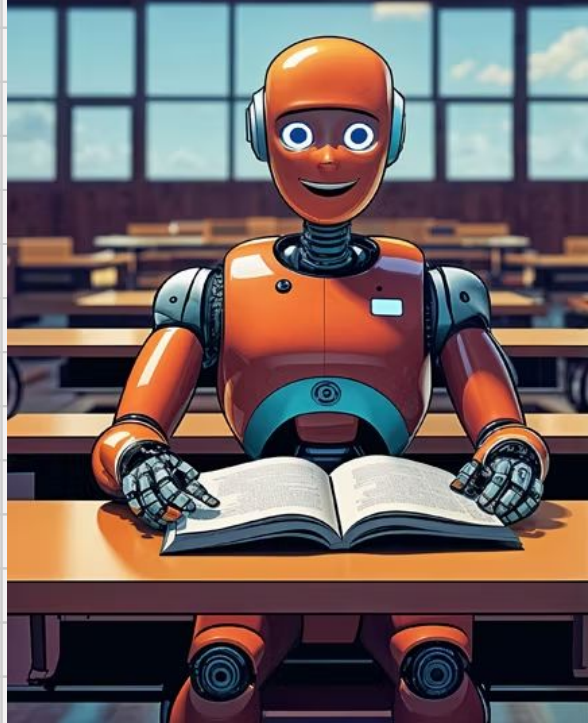




**PRE-TRAINING:
STUDY MORE**



**POST-TRAINING:
PRACTICE MORE**



**INFERENCE-TIME:
THINK MORE**



Announcements

NEWS

- Quiz #3 - 4/22; will cover Lectures 8-11
- Final project presentations
 - Slide content to include all work completed
 - Focus presentation on work done after last presentation

Final Presentations:

Date	4.10p - 4.30p	4.30p - 4.50p	4.50p - 5.10p
4/27	Team 6	Team 9	Team 10
4/29	Team 7	Team 2	Team 5
5/4	Team 3	Team 11	Team 4
5/6	Team 8	Team 1	



AI AND



ARTEMIS II



Join us as we:

- Learn about the Artemis II mission and how NASA used AI to aid astronauts
- Discuss the dangers of fake AI-generated space images online
- Watch the moon orbit!



Wednesday 4/15/26

12pm - 1pm

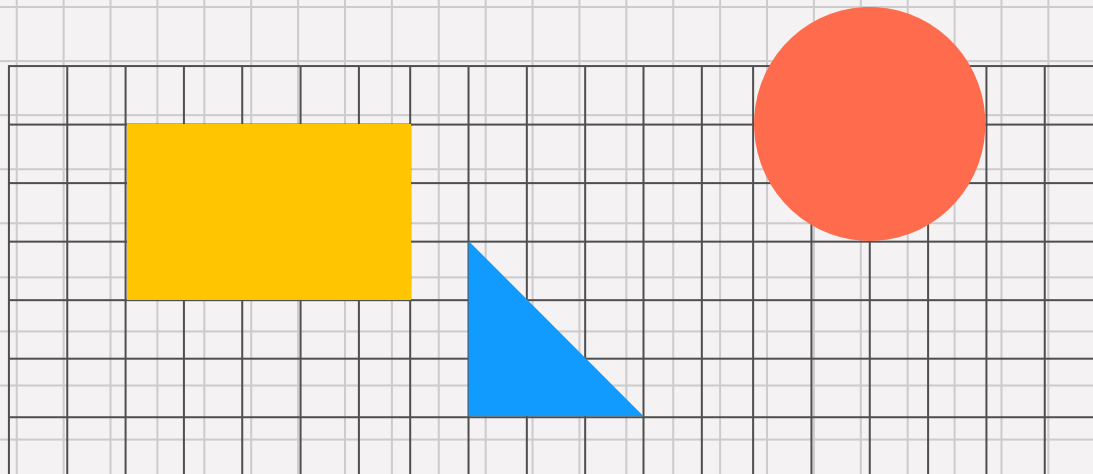
WRB 3004

****We will have pizza!**

Spring 2026

IS 365

Lecture 11: Reasoning in AI Evolving Role of Humans





Recap

Evolution of Agentic AI

01

Initial Conversational Agents

The first AI bots interacted with humans using natural language, but they had limited capabilities.

02

Generative AI

Generative AI emerged, improving the versatility and accessibility of AI.

03

Multimodal Advancement

AI has become multimodal, integrating multiple input and output formats.

04

Improved Reasoning

AI has improved its reasoning ability, handling complex tasks with less human intervention.

05

Greater Autonomy

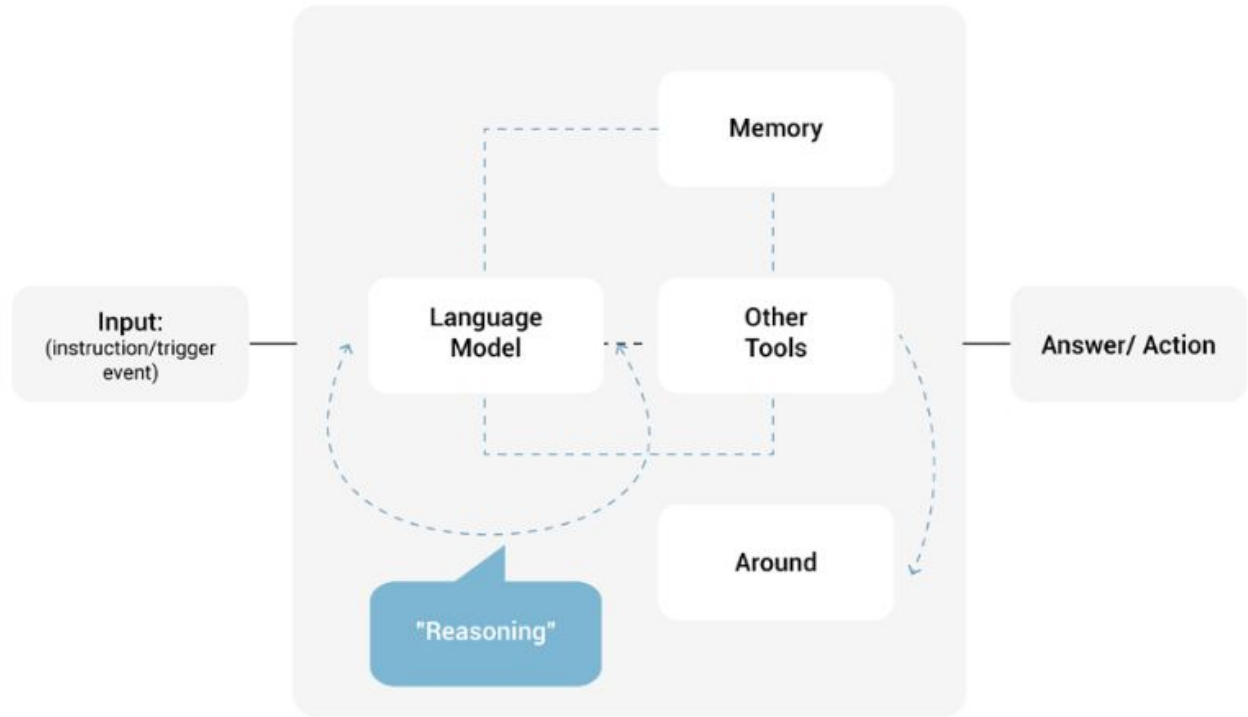
AI became more autonomous and adaptable, incorporating tools such as memory and internet access.

06

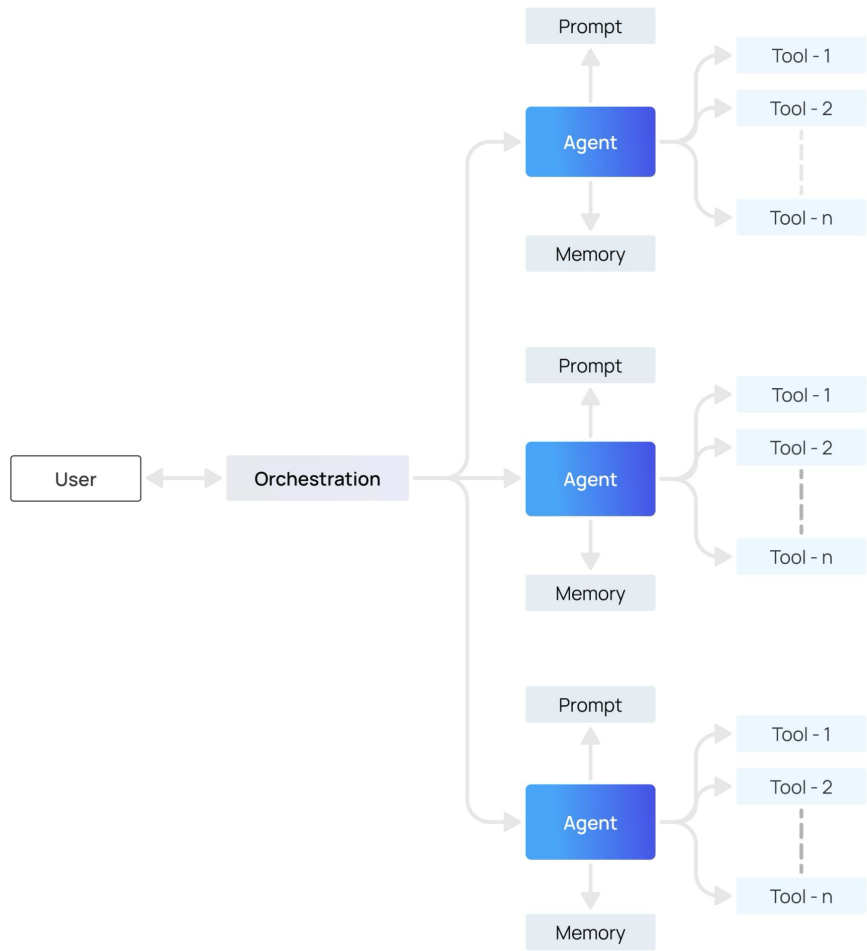
Specialized Agents

Highly specialized agents were developed, demonstrating high autonomy in specific tasks.

AI Agent

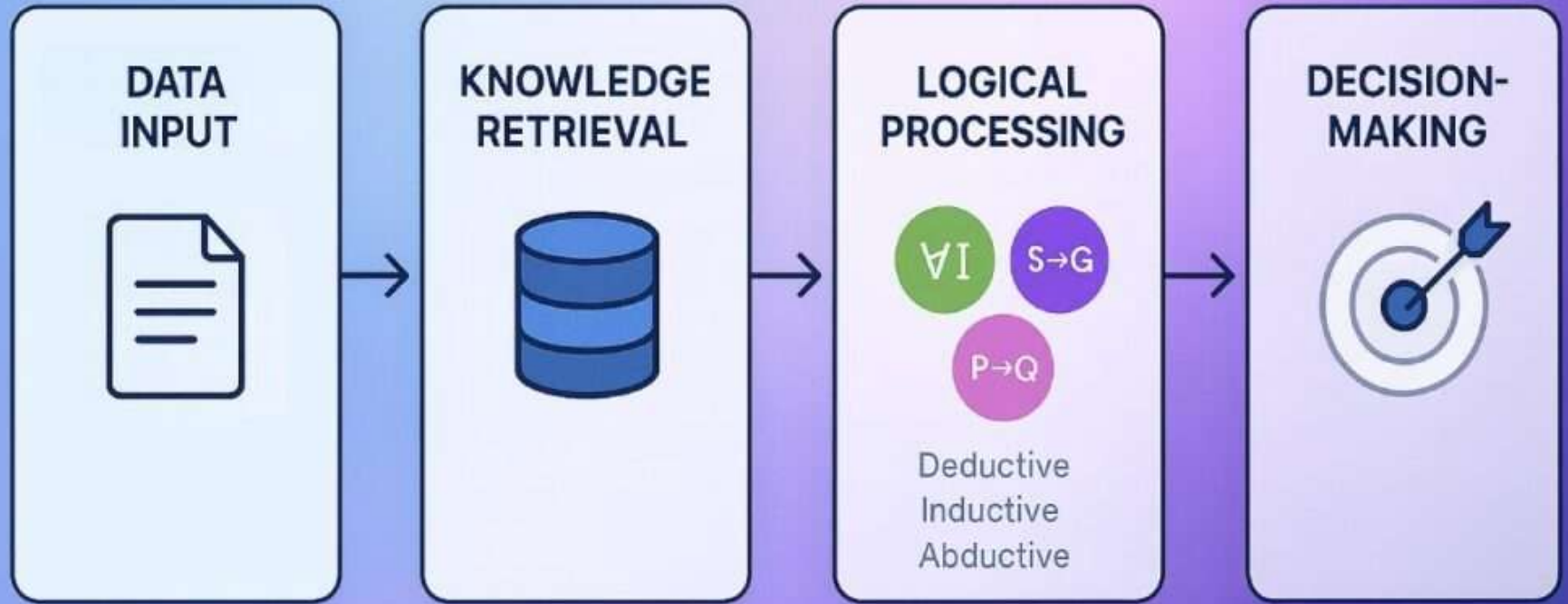


Agentic AI System



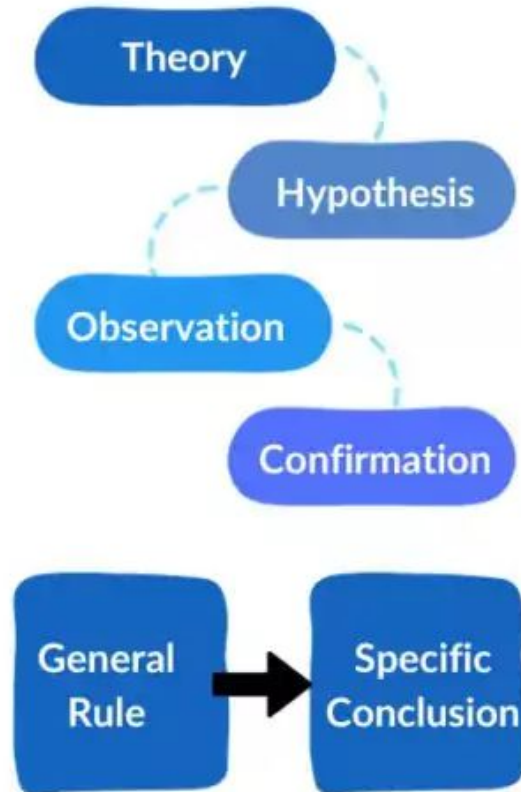
Types of Reasoning

AI REASONING SYSTEM





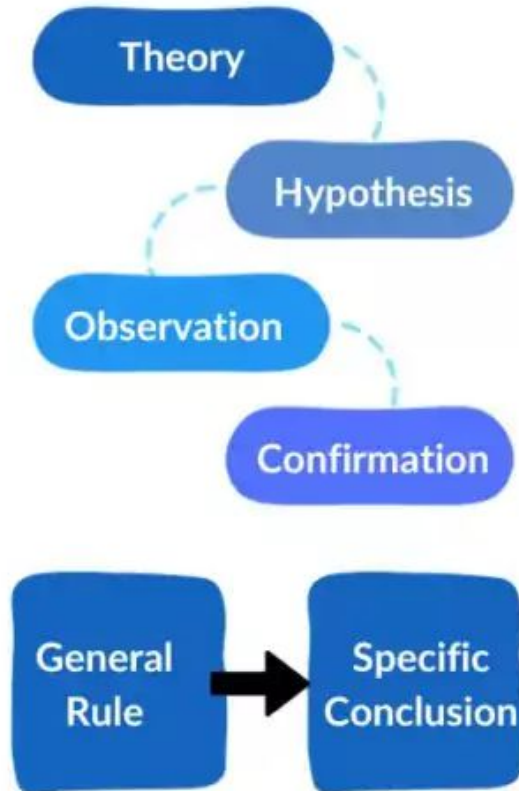
Deductive Reasoning



Deductive
All men are mortal;
Socrates is a man;
∴ Socrates is mortal.



Deductive Reasoning

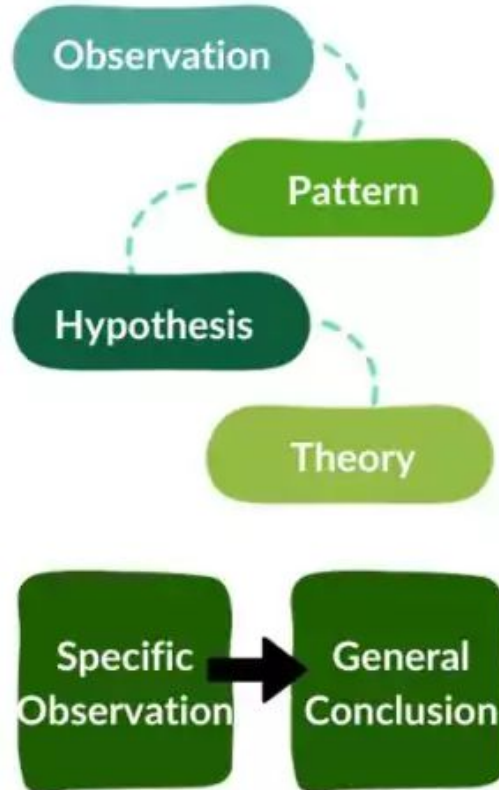


Enforcing business rules & compliance

Deductive
All men are mortal;
Socrates is a man;
∴ Socrates is mortal.



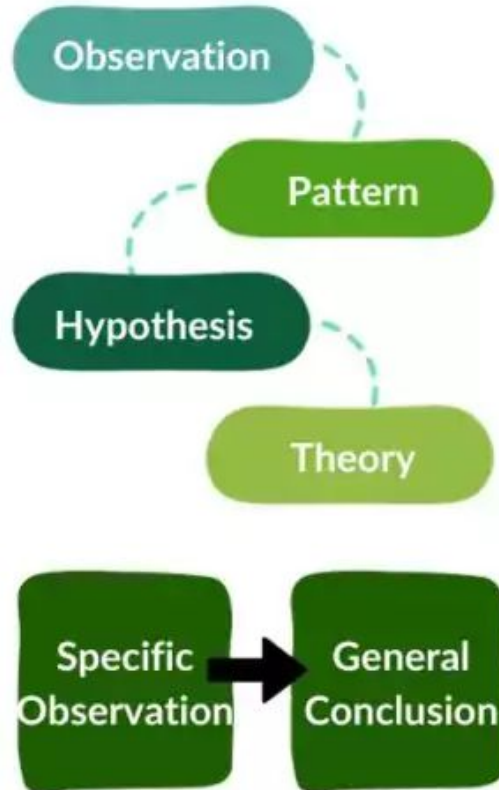
Inductive Reasoning



Inductive
Socrates is a man;
Socrates is mortal;
∴ All men are mortal.



Inductive Reasoning

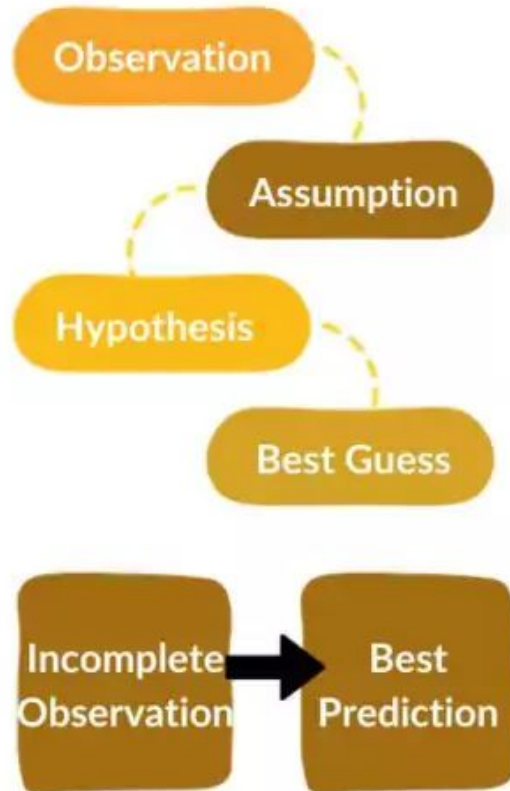


Predictive analytics & fraud detection

Inductive
Socrates is a man;
Socrates is mortal;
∴ All men are mortal.



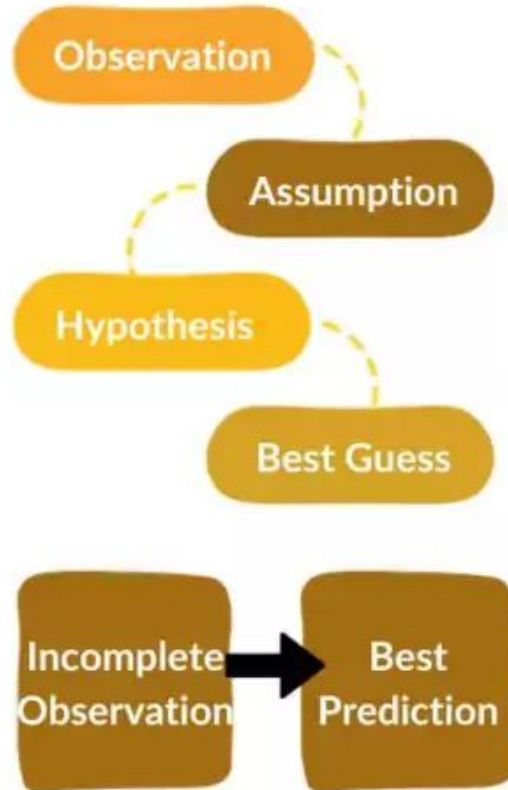
Abductive Reasoning



Abductive
All men are mortal;
Socrates is mortal;
$\therefore$ Socrates is a man.



Abductive Reasoning



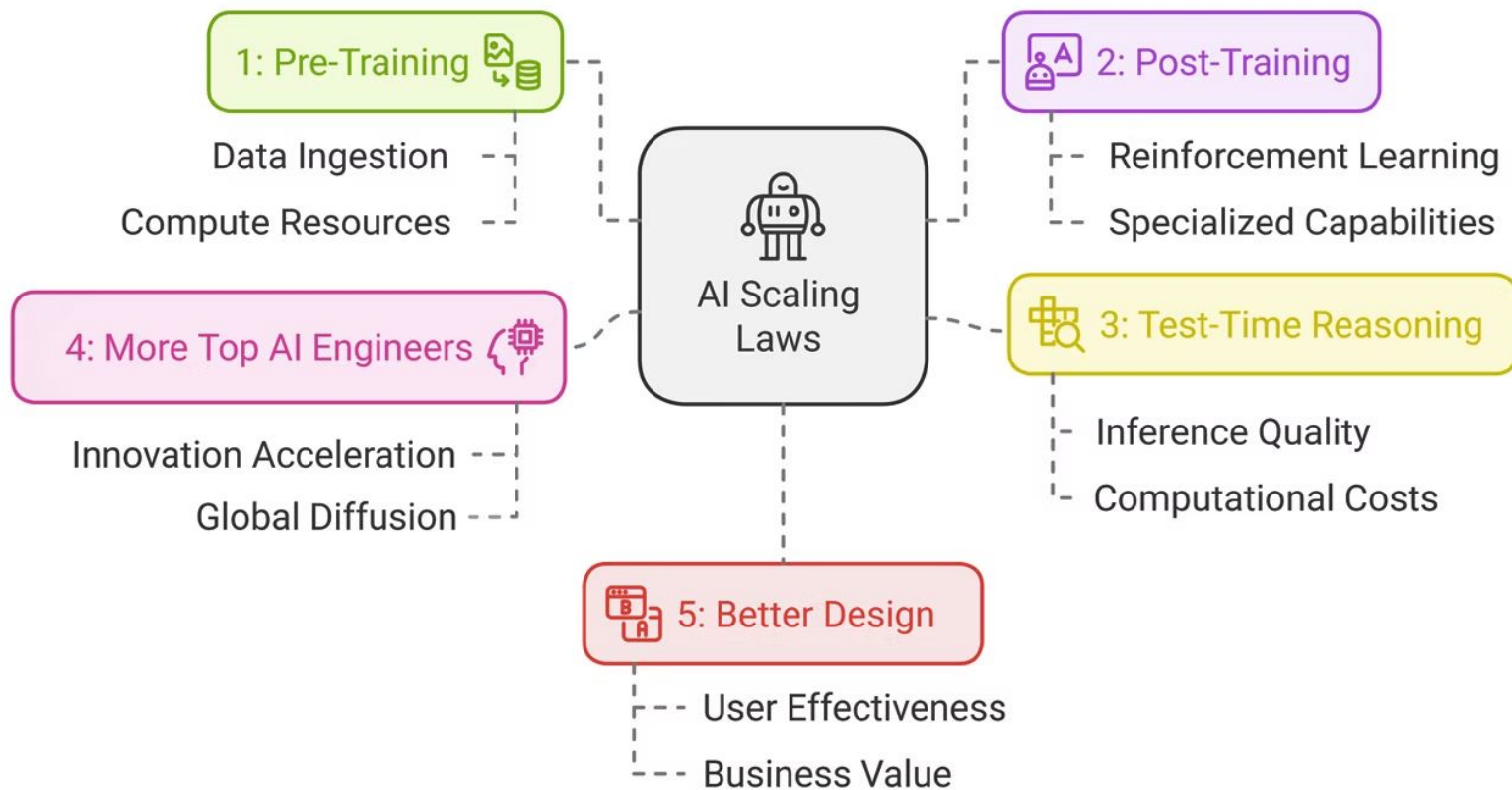
Medical diagnosis & troubleshooting

Abductive
All men are mortal;
Socrates is mortal;
$\therefore$ Socrates is a man.

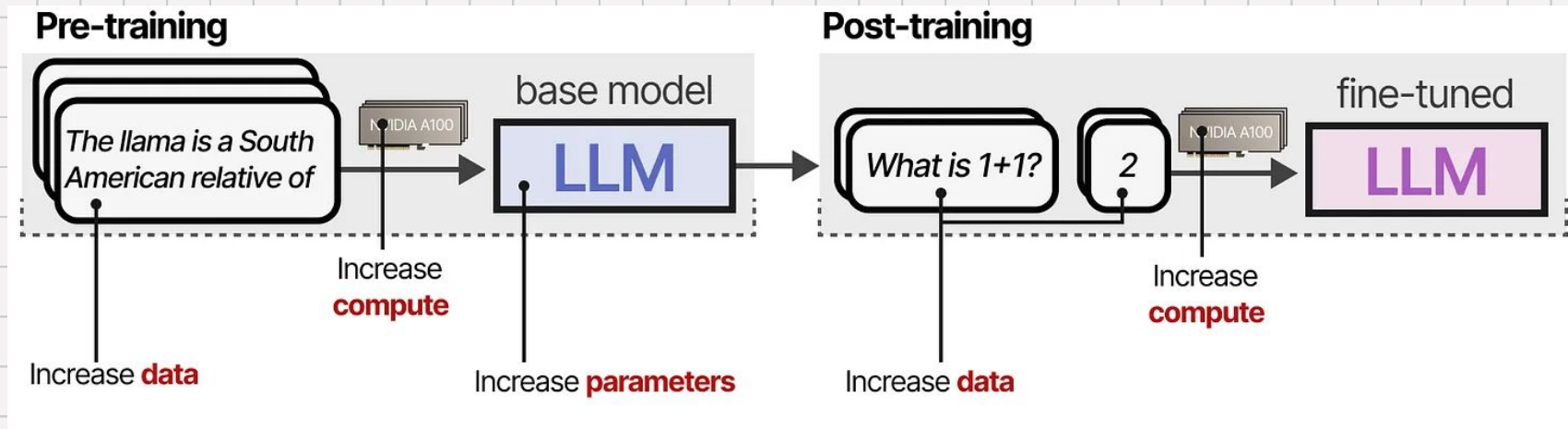
Reasoning Type	Description	Key Application
Analogical Reasoning	Solves new problems by drawing parallels to similar, previously solved problems.	Robotics, cognitive systems, and knowledge transfer.
Common Sense Reasoning	Simulates human everyday knowledge to handle nuanced or ambiguous situations.	Conversational AI and autonomous agents.
Monotonic Reasoning	Conclusions, once drawn, cannot be reversed even with new information.	Formal verification and conventional logic-based systems.
Nonmonotonic Reasoning	Allows for the revision of conclusions when new information becomes available.	Real-time traffic management and dynamic decision-making.
Fuzzy Reasoning	Handles uncertainty and imprecision by allowing degrees of truth rather than a binary outcome.	Control systems (e.g., in air conditioners, washing machines).

Need for Reasoning LLMs

The 5 AI Scaling Laws



Traditional GPT Models



Test-time compute
(FLOPS-seconds)

$$\log_{10}(\text{test}) = -1.2 \cdot \log_{10}(\text{train}) + 0.004 \cdot \text{elo} + 29$$

1e9

1e8

1e7

1e6

-1500

-1250

-1000

-750

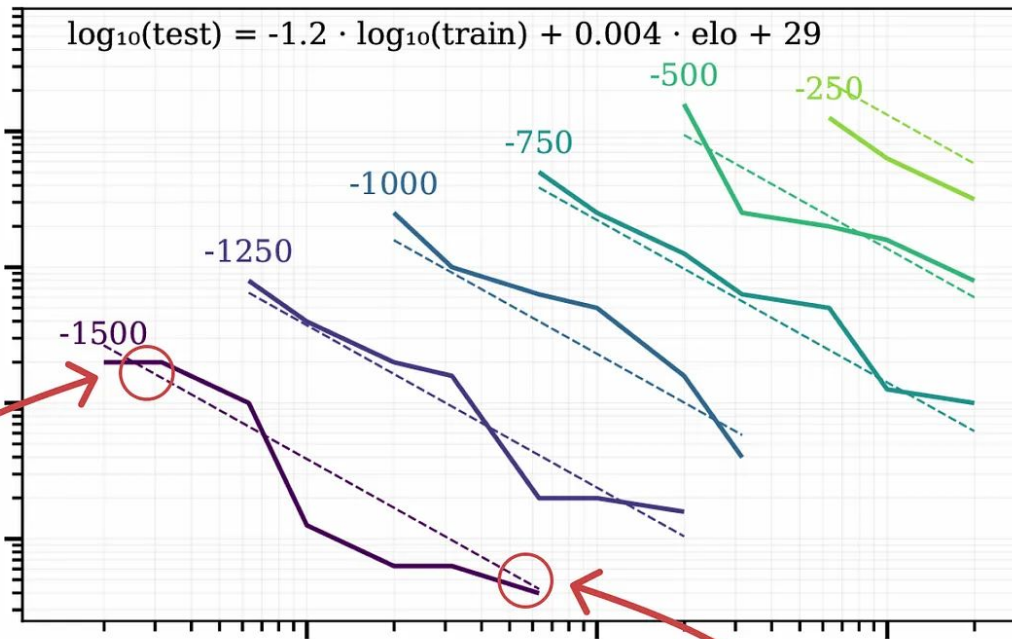
-500

-250

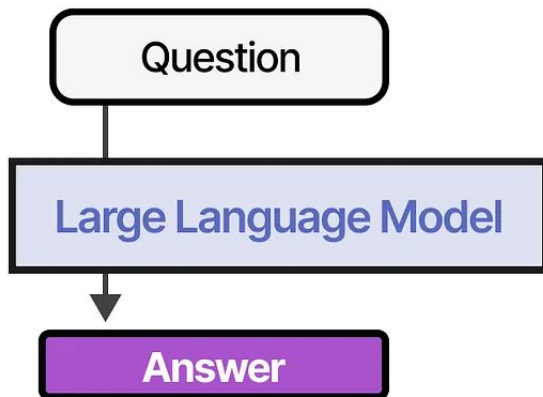
Train-time compute
(FLOPS-seconds)

More **test-time** compute
means less **train-time**
compute is needed.

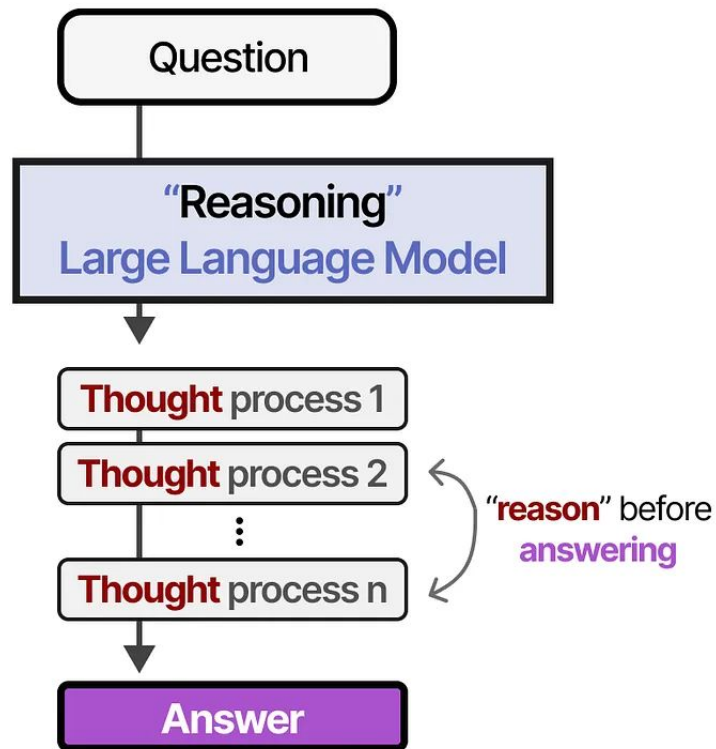
Less **test-time** compute
means more **train-time**
compute is needed.



“Regular” LLMs



“Reasoning” LLMs



Question

I have **10** apples. I gave **2** apples away. I ate **1**. How many do I have?

"Reasoning"
Large Language Model

You have **10** apples

You gave **2** away and have **8** left

You ate **1** and have **7** left

You have **7** apples

← **final answer**

reason steps
(typically Chain-of-Thought)

“Regular” LLMs

Pre-training

Fine-tuning

Inference

Train-time compute

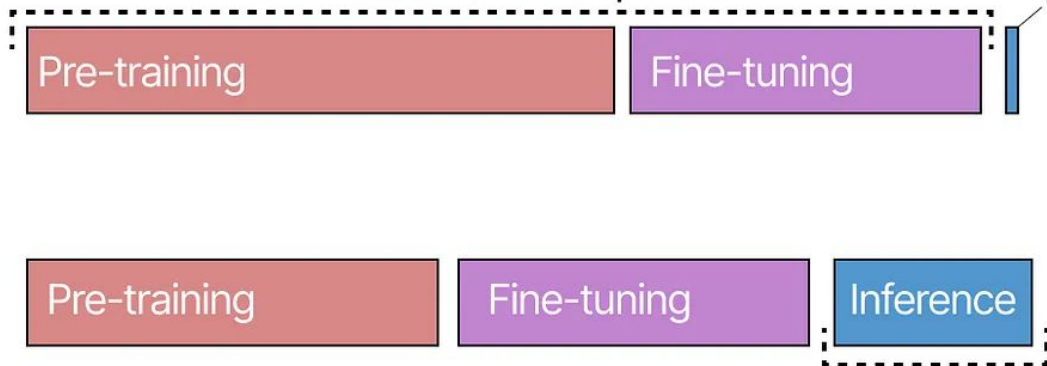
“Reasoning” LLMs

Pre-training

Fine-tuning

Inference

Test-time compute



OpenAI o1 Reasoning Model

OpenAI o1: New “Reasoning Paradigm”

- **OpenAI o1 marks a shift:** from 'generate' to 'reason'
- Traditional GPT models (*OpenAI GPTs, Gemini, Claude, Llama, Grok, etc.*) focus on **large-scale training** → bigger is better
- o1 emphasizes **reasoning and thinking** → time during inference

How o1 Reasons

- Uses **Chain-of-Thought (CoT) reasoning**: step-by-step problem solving
- Reinforcement Learning trains productive thinking strategies
- Model's chain of thought is hidden from users for **safety reasons**
- Reasoning mode uses more compute and time—cost trade-offs exist

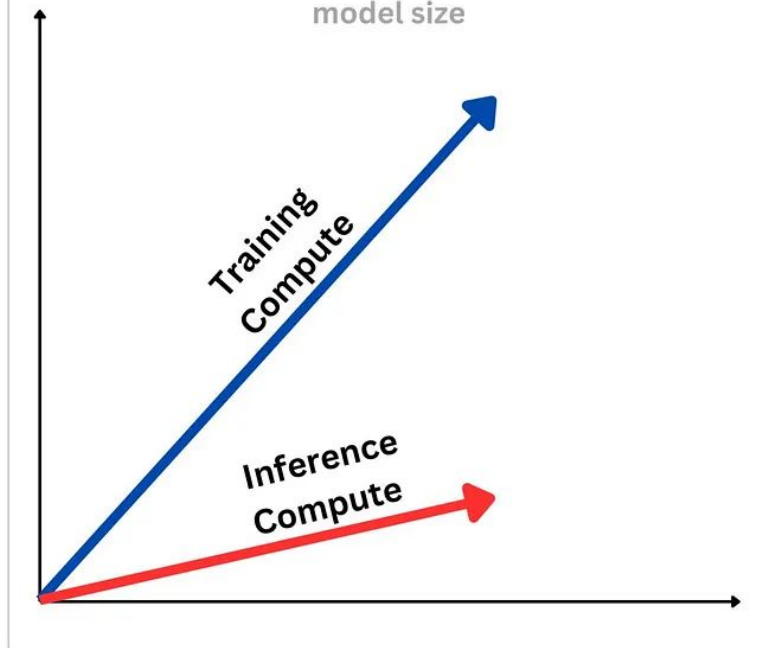
o1: Unique Differentiation

- Traditional LLMs allocate most compute to pre-training; inference was fast
- o1 allocates more compute at inference—allowing time to think
- Built-in reinforcement learning helps it self-evaluate reasoning
- Smaller model size can perform better with longer reasoning time



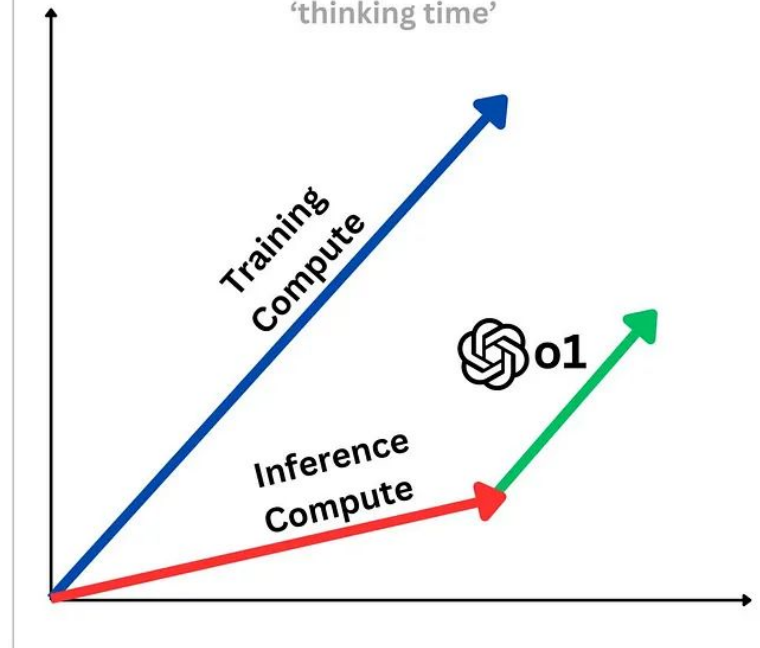
Before OpenAI o1

Inference compute scaled **linearly** with model size



After OpenAI o1

Inference compute scaled **exponentially** with 'thinking time'



o1: Benchmark Performance

- **AIME math benchmark:** o1 improves with more inference compute
- Excels on coding (Codeforces) and science benchmarks
- Reaches **89th percentile** on competitive programming tasks
- **Exceeds human PhD-level accuracy** on scientific questions
- Still **struggles** with simple reasoning puzzles

IQ Test Results

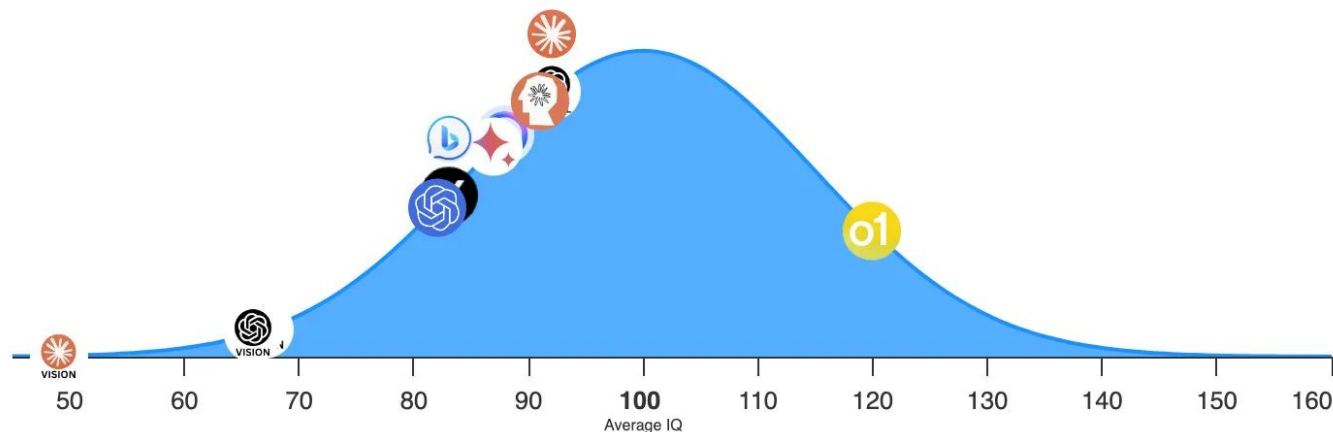
Reset

Show Offline Test

Show Mensa Norway



Score reflects average of last 7 tests given



OpenAI o1 preview



Gemini Advanced (Vision)



GPT4 Omni



Claude-3.5 Sonnet



Llama-3.1



Gemini Advanced



ChatGPT-4



Claude-3 Opus



Grok-2



GPT4 Omni (Vision)

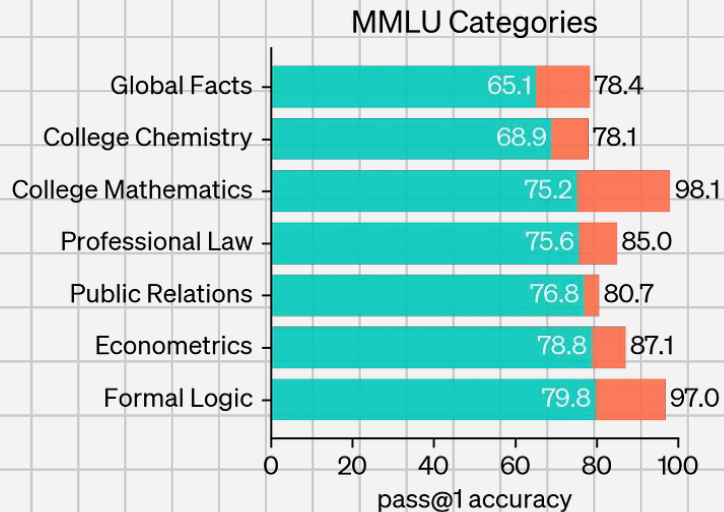
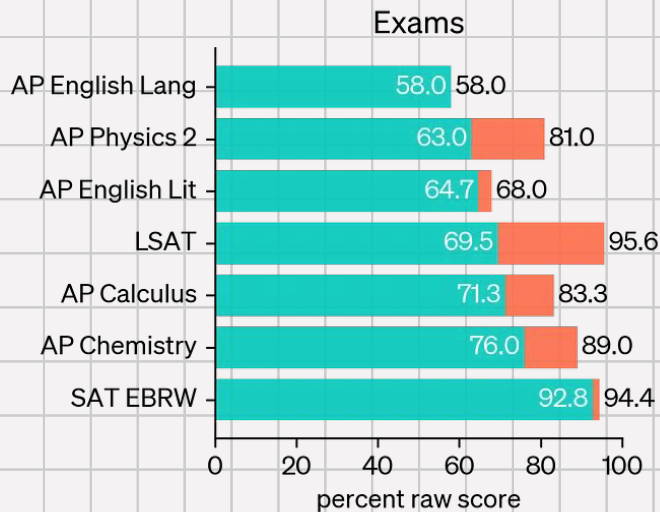
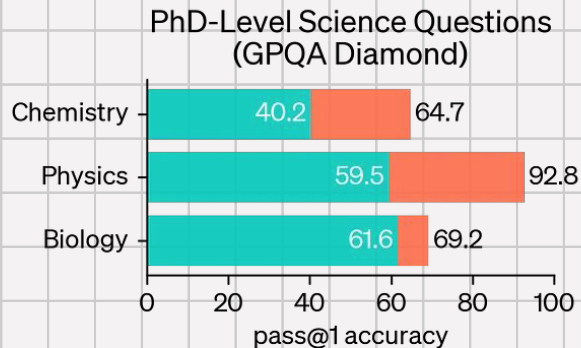
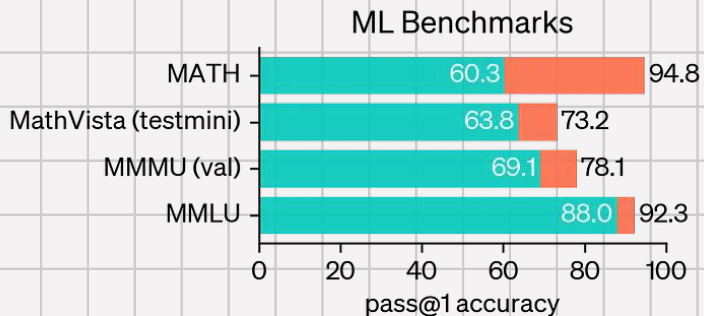


Bing Copilot



Claude-3 Opus (Vision)

gpt4o o1 improvement





Implications

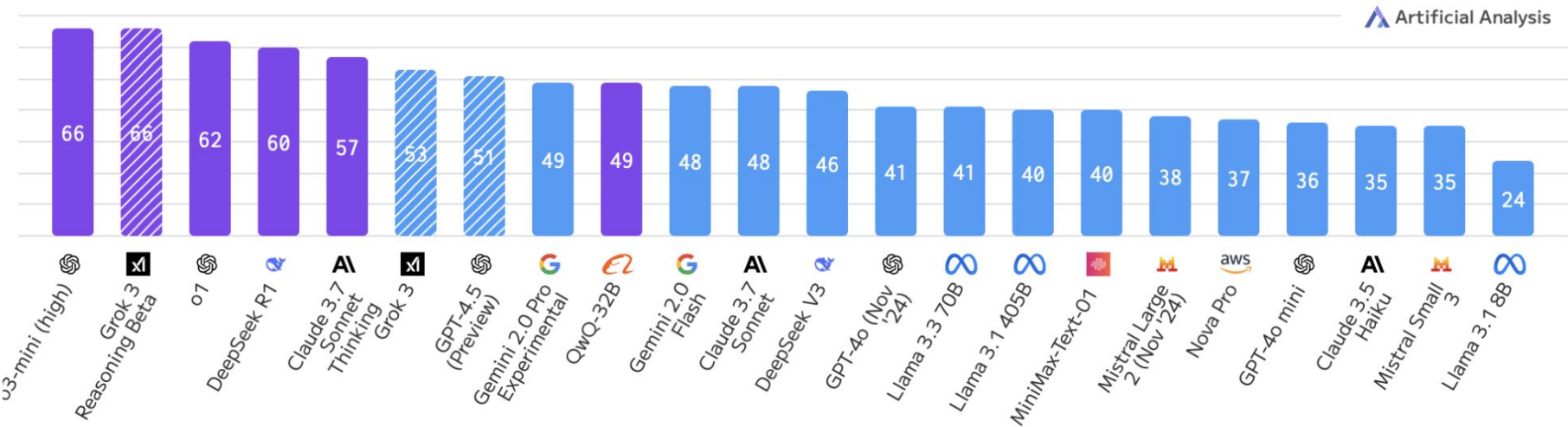
- **o1-mini**: smaller, cheaper version emphasizing reasoning efficiency.
- **GPT-style models** remain best for everyday general use
- **Reasoning models** target advanced domains (science, math, workflows)
- Issues:
 - Hidden reasoning raises transparency, ethical, and trust concerns
 - Questions around scaling cost, time, and access

Artificial Analysis Intelligence Index by Model Type

Intelligence Index incorporates 7 evaluations spanning reasoning, knowledge, math & coding

 Estimate (independent evaluation forthcoming)

 Reasoning Model  Non-Reasoning Model



Levels of Automation

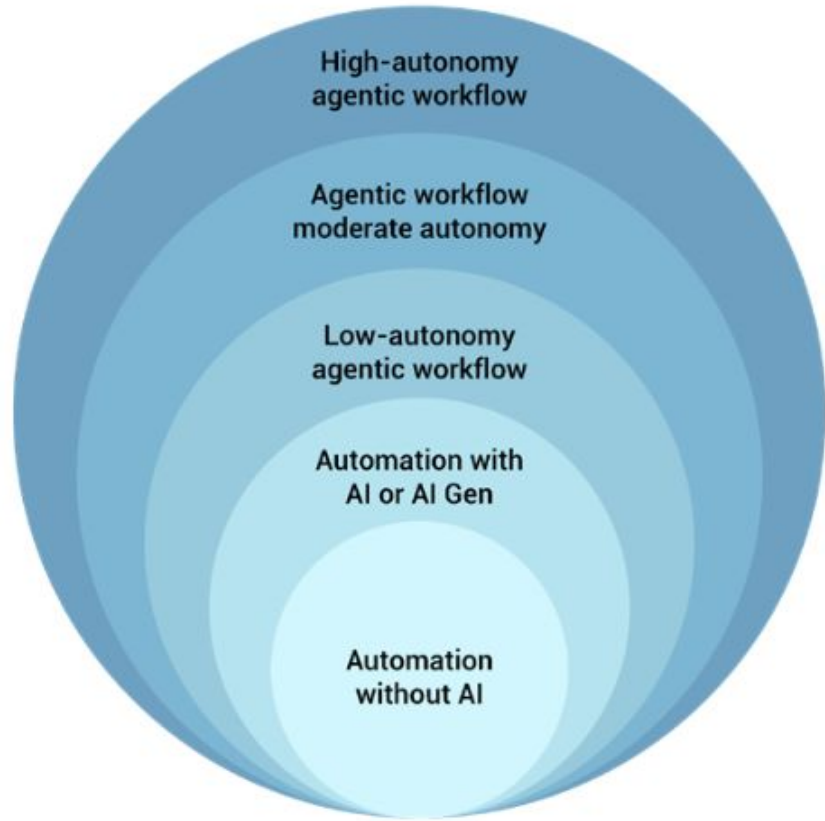
Choosing the Right Automation

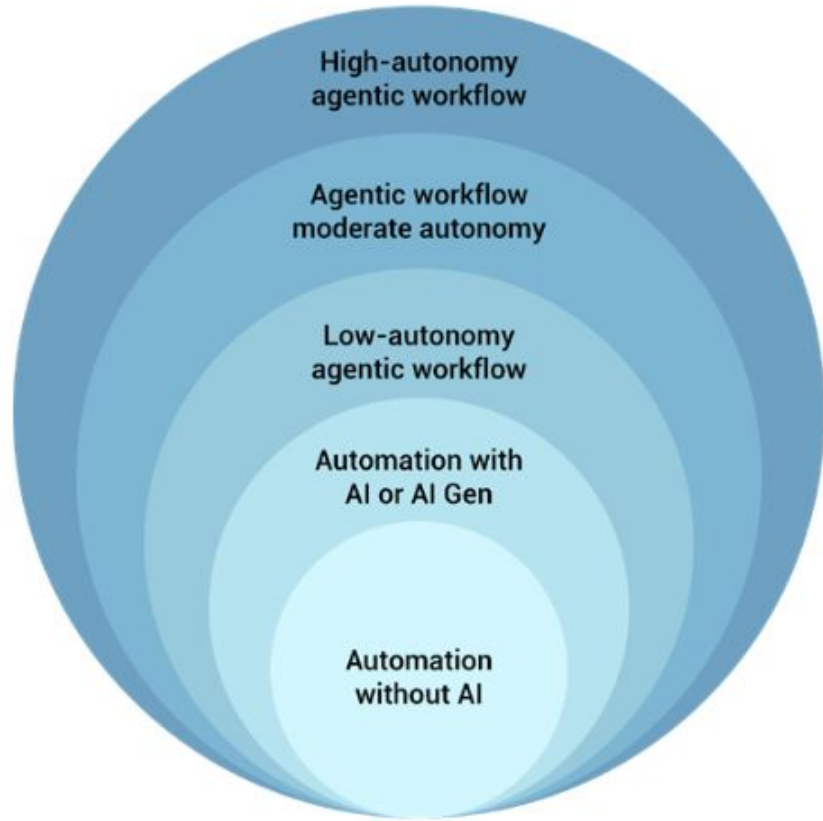
- Not all tasks require AI agents
- Simpler automations can be faster, cheaper, and safer
- AI agents suit dynamic, non-deterministic workflows
- Balance complexity with efficiency and risk

Automation Spectrum

- **Non-AI automation:** rule-based, deterministic workflows
- **AI-assisted workflows:** integrate LLMs via APIs
- **Agentic workflows:** autonomous, adaptive systems
- **Trade Offs:** reliability vs flexibility vs autonomy

Levels of Automation





Examples:

Level 1: Simple chatbot with fixed responses

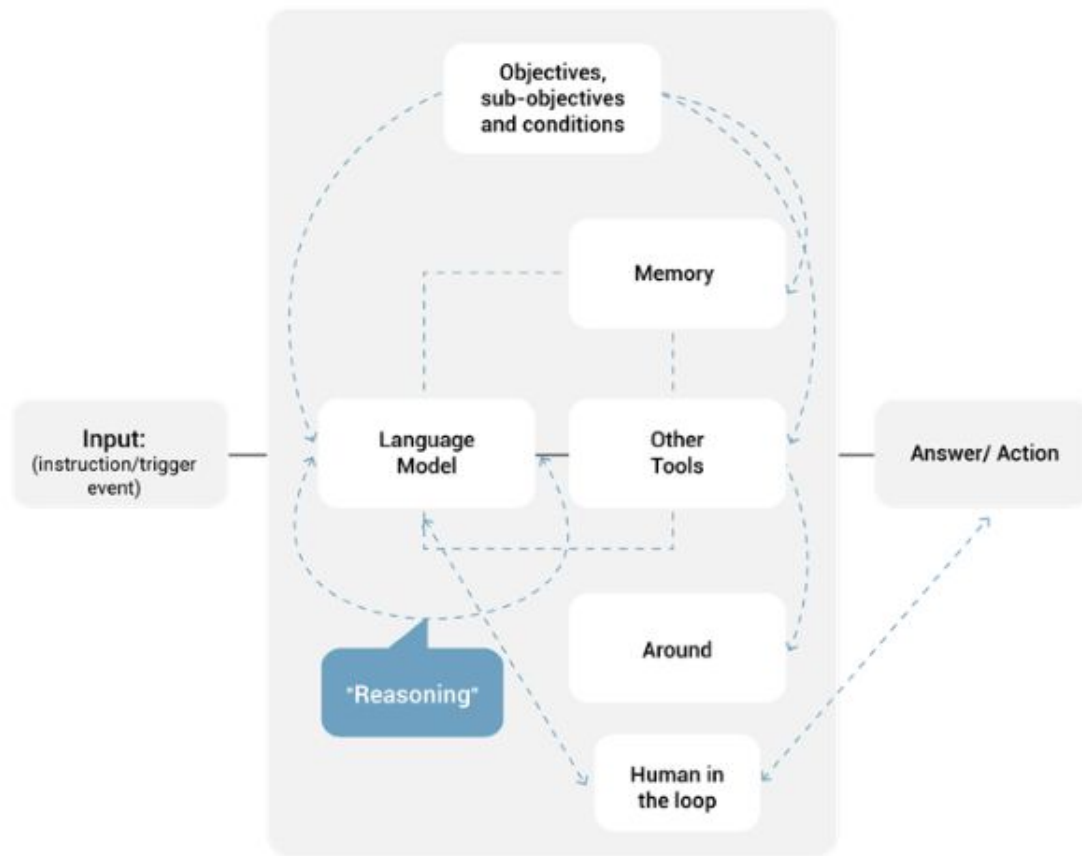
Level 2: ChatGPT summarizing input data

Level 3: Custom GPT automating legal document drafting

Level 4: Automated workflow combining APIs and AI

Level 5: Fully autonomous trip planning agent

Human Augmentation



Human-Augmented Agents

- Human oversight remains critical in AI workflows
- Probabilistic outputs can cause errors or biases
- Humans validate and refine system outputs
- Human-in-the-loop ensures accountability and ethics

Why Human Control Matters

- AI hallucinations and unpredictability require supervision
- Human judgment prevents harmful or illegal outputs
- Critical in fields like law, healthcare, and governance
- Reinforces trust and reliability in AI systems

Human-in-the-Loop Design

- Humans **intervene** at key workflow checkpoints
- Validate, replace, or cancel AI outputs as needed
- Creates Human Augmented Agents (HAA)
- Combines machine efficiency with human ethics

01

Validate, refine or override AI-generated responses.

02

Mitigate errors and biases before deployment to production.

03

Building trust through collaborative validation processes.

04

Prevent hallucinations and ensure regulatory and ethical compliance.

05

Improve system learning through human feedback.

The balance between autonomy and human control is key to designing trustworthy agents aligned with ethical and organizational values.

Ethical Considerations

- Transparency and explainability are essential
- Maintain human oversight and accountability
- Design for fairness, privacy, and inclusivity
- Implement governance frameworks for agentic systems

Summary

- AI agents blend reasoning, adaptability, and collaboration
- Automation scales from rule-based to autonomous systems
- Human role shifts to oversight and quality assurance
- Agentic workflows redefine productivity and decision-making